Graph on the Coordinate Plane

Lesson 9-1

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A grid with a horizontal line (x-axis) crossing a vertical line (y-axis), making four sections

Coordinate plane

A grid with a line going across and a line going up that cross.

(3, 5) means move right 3 from the origin, then up 5

Ordered pair

Two numbers (x, y) that tell where a point is on a grid.

The starting point at the center where both axes cross — (0, 0)

Origin

The point (0, 0) where the two grid lines cross.

I (+,+) top-right, II (-,+) top-left, III (-,-) bottom-left, IV (+,-) bottom-right

Quadrant

One of the four parts of a coordinate grid.

(3, 2) reflected over the x-axis becomes (3, -2) — same x, opposite y

Reflection

A flipped, mirror image across a line.

..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Whole numbers and their opposites, like -2, -1, 0, 1, 2.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can plot and identify points on the coordinate plane using ordered pairs.

1️⃣ Notice

Captain Vega left a treasure map, but instead of an 'X marks the spot,' she hid the directions as ordered pairs!

2️⃣ Key idea

I can plot and identify points on the coordinate plane using ordered pairs.

3️⃣ Words I'll use

Coordinate plane **Ordered pair** **Origin** **Quadrant** **Reflection** **Integer**

🤔 Think About It

- What shape will the three points make when connected?
- Which number tells you how far right/left? Which tells you up/down?
- Why is the ORDER in an ordered pair important?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: Which ordered pair names a point 3 units right and 7 units up from the origin?

A. (3, 7)
B. (7, 3)
C. (3, 3)
D. (0, 7)

Step 1 The first coordinate is horizontal (right 3), the second is vertical (up 7).

Step 2 So (3, 7).

✅ **Answer:** A. (3, 7)

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: What are the coordinates of the origin?

A. (0, 0)

B. (1, 1)

C. (0, 1)

D. (1, 0)

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: A point is at (6, 2). What does the 6 tell you?

A. Move 6 units to the right

B. Move 6 units up

C. Move 6 units to the left

D. The point is in Quadrant 6

Show your work:

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

On the coordinate plane, why do we always start at the origin (0, 0) and move sideways before we move up or down to plot a point?

Level 1 support

Start here: We start at the origin and move along the x-axis first because ____.

💡 **Need a hint?**

Sentence starters (optional)

The first number in (x, y) tells me to move ____.
El primer numero en (x, y) me dice que me mueva ____.

The second number tells me to move ____.
El segundo numero me dice que me mueva ____.

Word bank: coordinate plane | origin | ordered pair | x-axis | y-axis

Listen for: *Student states the x-coordinate is the sideways (horizontal) move and the y-coordinate is the up/down (vertical) move, both starting from the origin.*

Level 2

Push students to explain why (3, 5) and (5, 3) land on different points even though they use the same numbers.

- (3, 5) and (5, 3) are different because ____.
- The order in an ordered pair matters because ____.

EXPLORE

You need to plot (4, 2). Walk your partner through every move you make from the origin, using the words x-axis and y-axis.

Level 1 support

Start here: Starting at the origin, I first move ___, then I move ___.

💡 **Need a hint?**

Sentence starters (optional)

I move ___ units along the x-axis.
Me muevo ___ unidades por el eje x.

Then I move ___ units along the y-axis.
Luego me muevo ___ unidades por el eje y.

Word bank: origin | ordered pair | x-axis | y-axis | coordinate plane

Listen for: *Student moves 4 right along the x-axis and 2 up along the y-axis, naming each axis and the origin as the start.*

Level 2

Ask students to predict which quadrant the point (4, 2) lands in and justify their answer using the signs of the coordinates.

- (4, 2) lands in quadrant ___ because ___.
- I can tell the quadrant from the coordinates by ___.

CONNECT

How is reading a coordinate pair like reading directions on a map or in a video game?

Level 1 support

Start here: An ordered pair works like map directions because ____.

💡 **Need a hint?**

Sentence starters (optional)

The x-coordinate is like going ____.

La coordenada x es como ir ____.

The y-coordinate is like going ____.

La coordenada y es como ir ____.

Word bank: ordered pair | coordinate plane | x-axis | y-axis | origin

Listen for: *Student connects the x-value to horizontal travel and the y-value to vertical travel, like map or grid directions from a starting point.*

Level 2

Push students to generalize a rule for plotting ANY ordered pair, including ones with zero in them.

- A rule that always works for plotting (x, y) is ____.
- If one coordinate is 0, the point lands on ____ because ____.

REFLECT

A classmate plotted (2, 6) but moved up 2 and over 6. What mistake did they make, and how would you coach them?

Level 1 support

Start here: The mistake was ____, and to fix it they should ____.

💡 **Need a hint?**

Sentence starters (optional)

They mixed up the ____ and the ____.

Confundieron la ____ y la ____.

The correct way is to move ____ first.

La forma correcta es moverse ____ primero.

Word bank: ordered pair | x-axis | y-axis | origin | coordinate plane

Listen for: *Student identifies that the x- and y-coordinates were swapped and explains the correct order (x first, then y).*

Level 2

Ask students to critique why this swap error is so common and suggest a strategy to avoid it every time.

- This error happens a lot because ____.
- A strategy to remember the order is ____.

Write About the Math

The Writing Revolution

I can explain my work using the words coordinate plane, ordered pair, origin, and quadrant.

1. Kernel Sentence

subject + verb

Model: Coordinate plane is a grid with a line going across and a line going up that cross.
Plano cartesiano es una cuadrícula con una línea horizontal y una vertical que se cruzan.

Write a kernel sentence about coordinate plane. Use a subject and a verb.
Escribe una oración base sobre plano cartesiano. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Coordinate plane matters in math
Plano cartesiano importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*

Coordinate plane matters in math because ____.
Plano cartesiano importa en matemáticas porque ____.

____

but *pero*

Coordinate plane matters in math, but ____.
Plano cartesiano importa en matemáticas, pero ____.

____

so *entonces*

Coordinate plane matters in math, so ____.
Plano cartesiano importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about coordinate plane.
Di un hecho verdadero sobre coordinate plane.

Coordinate plane ____.

____

Question
Pregunta

Ask a question about coordinate plane.
Haz una pregunta sobre coordinate plane.

How does ____ ?
¿Cómo ____ ?

____

Exclamation
Exclamación

Show excitement about coordinate plane.
Muestra entusiasmo sobre coordinate plane.

Wow, ____ !
¡Guau, ____ !

____

Command
Mandato

Tell a partner what to do with coordinate plane.
Dile a un compañero qué hacer con coordinate plane.

First, ____ .
Primero, ____ .

____

4. Explain Your Reasoning

use a sentence starter

The first number in (x, y) tells me to move ____.
El primer numero en (x, y) me dice que me mueva ____.

The second number tells me to move ____.
El segundo numero me dice que me mueva ____.

I move ____ units along the x-axis.
Me muevo ____ unidades por el eje x.

____

____

____

Try It

Solve on your own. Check the answer key when you are done.

1. A point lies on the x-axis at x = -6. What is its y-coordinate?

A. 0

B. -6

C. 6

D. 1

Show your work:

2. In which quadrant is the point (7, -1)?

A. Quadrant IV

B. Quadrant I

C. Quadrant II

D. Quadrant III

Show your work:

Stretch Your Thinking `Level 2 enrichment`

Challenge task — explain your reasoning in full sentences.

A student says that (3, 5) and (5, 3) are the same point because they use the same numbers. Is the student correct? Explain why or why not, and describe where each point is located on the coordinate plane.

Sentence starter: The student is ____ because (3, 5) means ____ while (5, 3) means ____. The ORDER matters because ____.

Show your work:

____

____

____

____

Reflect — Exit Ticket

Point P is located 6 units right and 3 units up from the origin. What ordered pair names point P?

A. (6, 3)
B. (3, 6)
C. (6, 0)
D. (0, 3)

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **We Try (notes):** A. (0, 0) *— The origin is where the x-axis and y-axis cross, at (0, 0).*
2. **You Try (notes):** A. Move 6 units to the right *— The first number (x-coordinate) tells you horizontal movement. 6 means go right 6 units from the origin.*
3. **Try It 1:** A. 0 *— Points on the x-axis always have y = 0, so the point is (-6, 0).*
4. **Try It 2:** A. Quadrant IV *— Positive x and negative y place a point in Quadrant IV.*
5. **Exit Ticket:** A. (6, 3) *— Right 6 = x-coordinate of 6, Up 3 = y-coordinate of 3. The ordered pair is (6, 3).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Coordinate plane is a grid with a line going across and a line going up that cross.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).